

Hespeler Park, Niverville MB

Drainage Assessment

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ACKNOWLEDGEMENT AND DISCLAIMER

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ABSTRACT

This project uses geomatics methods to identify, map, and interpret flooding and drainage problems in Hespeler Park, Niverville, Manitoba. The main objective is to create a repeatable flood-zone inventory that combines field GPS observations, aerial imagery, LiDAR-derived elevation data, and hillshade interpretation within a structured QGIS database workflow. Field observations were imported into QGIS and saved as permanent GeoPackage layers using EPSG:26914, allowing mapped flood-prone areas to be classified, updated, and compared over time. A LiDAR-derived DEM was used as the main elevation surface, and hillshade layers were generated to improve interpretation of subtle terrain variation and drainage patterns. A flood polygon layer was then created to digitize and classify flood-prone zones according to severity, confidence, likely cause, drainage conditions, and priority. The broader goal is to test whether open-source GIS software, open elevation data, field observations, and structured database design can provide a low-cost method for documenting localized flooding and supporting future drainage assessment before more expensive engineering interventions are considered.

INTRODUCTION AND OBJECTIVES

The goal of this project is to document flood-prone areas within Hespeler Park using consistent and repeatable geomatics methods. The analysis focuses on identifying, classifying, and interpreting flood-prone terrain through a structured polygon inventory supported by field observations, aerial imagery, and LiDAR-derived elevation data. Subtle surface patterns were interpreted by comparing elevation, relief, and imagery, then recorded in a permanent spatial database so they can be measured, classified, and updated over time.

Elevation is central to the interpretation because local lows, poor drainage, and terrain position help explain why some areas dry seasonally while others flood repeatedly or remain wet for longer periods. By combining raster-based terrain analysis with vector-based mapping, the project produces a flood-zone inventory that can support comparison, prioritization, and future reassessment if site conditions change.

The project applies a database-driven geomatics workflow to a practical local drainage problem. Rather than treating flooding as isolated observations, the workflow organizes field evidence, terrain interpretation, and mapped flood zones into a structured GIS database. This allows flood-prone areas to be documented in a consistent format and reviewed again after future rainfall, snowmelt, maintenance, or mitigation work.

The broader objective is to assess whether this workflow could become a repeatable, low-cost method for documenting and monitoring localized flooding. The project also identifies what additional data, monitoring, modelling, or professional review would be needed to bring the process closer to a professional drainage assessment standard.

METHODS

This project was developed as a GIS-based flood assessment workflow for Hespeler Park in Niverville, Manitoba. The workflow combines field observations, aerial imagery, LiDAR elevation data, and terrain visualization to identify, classify, and document localized flooding and drainage concerns.

The result is a structured flood-zone inventory built in QGIS. Mapped flood-prone areas were organized in a spatial database and classified by severity, confidence, drainage condition, likely cause, visibility, and priority. This structure allows flood areas to be reviewed, updated, compared, and tracked over time.

Terrain assessment was completed using LiDAR-derived elevation data, hillshade visualization, aerial imagery, and GPS-based field observations. These datasets were used to identify low-lying terrain, visible wet areas, snowmelt pooling, drainage interruptions, and possible surface-flow patterns.

Flood zones were digitized as polygons and assigned classification values during mapping. Additional derived layers and spatial summaries were used to compare flood-zone area, elevation, priority, and distribution across the park.

The workflow is designed as a low-cost GIS framework for documenting drainage issues, organizing field evidence, and supporting future maintenance or mitigation planning.

RESULTS

A clipped digital elevation model was produced for the Hespeler Park study area to isolate the terrain surface used for analysis (see Appendix A). From this clipped DEM, both grey and coloured hill shade layers were generated, along with a coloured DEM layer, to improve interpretation of subtle relief, low terrain, and potential drainage pathways (see Appendix B, Appendix C, and Appendix D).

A structured flood polygon dataset was created within a GeoPackage, including fields for classification, confidence, drainage conditions, and likely causes of flooding (see Appendix E).

Flood-prone areas were digitized as polygons from the observations and stored in the Hespeler Flood Zones layer, producing a mapped inventory of observed and inferred flooding across the study area (see Appendix F1 and F2).

The flood polygon dataset includes structured attribute information for each feature, allowing classification and comparison of flood severity, confidence, and likely causes (see Appendix G).

Summary statistics of the Hespeler Flood Zones dataset indicate variation in both frequency and spatial extent across flood tiers. TIER_2 zones dominate both in count ($n = 27$) and total area ($113,122 \text{ m}^2$), while TIER_1 zones are smaller and less extensive ($n = 8$; $10,218 \text{ m}^2$). TIER_3 zones, although limited in number ($n = 4$), exhibit the largest mean area ($5,235 \text{ m}^2$) and highest minimum extents ($2,545 \text{ m}^2$), indicating that the most severe flooding occurs in fewer but more spatially continuous areas which was expected (see Appendix H).

Nearest neighbour distance analysis of flood zone centroids showed that mapped flood areas are generally located close to one another rather than being evenly distributed across the park. Distances ranged from 29.73 to 188.04 m, with a mean of 64.05 m, a median of 55.22 m, and a standard deviation of 31.38 m. The lower median relative to the mean indicates that many flood zones occur in proximity, while a smaller number of more isolated zones increase the overall average distance (see Appendix I).

Global Moran's I analysis of flood tier values showed weak positive spatial autocorrelation ($I = 0.1300$), which means a slight tendency for similar flood severity levels to occur near one another. However, this pattern was not statistically significant ($p = 0.1229$), suggesting that the spatial distribution of flood severity across Hespeler Park does not differ significantly from a random pattern (see Appendix J).

Flood polygon area varied across severity tiers, with higher-tier flood zones generally occupying larger areas. Tier 3 flood zones showed the greatest average area, while Tier 1 zones were smaller and more localized. The spread of values within each tier indicates variability in flood extent across the park, with some larger outliers present (see Appendix K).

Flood zones were categorized into priority levels based on observed severity and drainage limitations. The majority of the mapped flood zones fall within the medium priority category, with fewer zones classified as high or low priority. This distribution reflects a predominance of moderate drainage issues across Hespeler Park, with a smaller number of locations requiring more urgent intervention (see Appendix L).

All statistical analysis and spatial pattern calculations were conducted in R. The cleaned code used for the final analysis is provided in Appendix M1 and Appendix M2.

Kernel density analysis of flood zone centroids shows clear spatial clustering of flooding within localized areas of Hespeler Park. High-density zones indicate concentrated groupings of flood-prone areas, while low-density regions show minimal or isolated flooding (see Appendix N). These patterns reinforce the nearest neighbour results, confirming that flooding is not evenly distributed but occurs in distinct clusters across the study area.

Ripley's K-function analysis of flood zone centroids indicates clustering at short to moderate distance ranges, where observed values exceed those expected under a random distribution. This suggests that flood-prone areas occur in localized groupings across the study area. At larger distances, the pattern approaches randomness, indicating that clustering is not consistent across the entire park but is instead concentrated within specific areas (see Appendix O).

An aspect-derived terrain direction map was produced from the filled DEM to represent the direction of local slope across the study area (see Appendix Q). When compared with hill shade and mapped flood zones, the aspect layer shows that many flood-prone areas occur where surrounding slope directions converge toward local low-lying terrain. This pattern supports the interpretation that surface water is being directed into these areas by small-scale variations in elevation and slope direction.

DISCUSSION

The objective of this project was to develop a repeatable geomatics workflow for identifying, classifying, and documenting flood-prone areas within Hespeler Park. The main result is not simply a map of wet areas. It is a structured GIS inventory that can be used to track drainage problems over time. Flood-prone areas were mapped, classified, and stored in a spatial database so that each location can be reviewed, updated, compared, and reassessed after future rainfall, snowmelt, maintenance, or mitigation work.

The value of the workflow is that it turns field observations and terrain interpretation into organized spatial information. Instead of treating drainage issues as separate observations, the database records each mapped area by severity, confidence, visibility, drainage condition, likely cause, and priority. This makes the information easier to retrieve and compare, while also creating a practical framework for monitoring whether drainage problems stay the same, worsen, improve, or shift over time.

The low-cost part of the workflow comes from its use of open-source GIS software, available elevation data, aerial imagery, and repeatable field observations. The main ongoing cost would not be specialized software, but the time required for field checks, database updates, interpretation, and maintenance of the GIS project. This makes the workflow useful as a monitoring framework because the database can grow as new observations, rainfall events, snowmelt conditions, or maintenance work are added.

The combination of field observations, aerial imagery, LiDAR-derived elevation data, and hillshade interpretation provided a consistent basis for identifying low-relief depressions, wet areas, and poorly drained locations. This is important in a low-relief landscape such as the Red River Valley, where small elevation differences can strongly influence whether water drains, pools temporarily, or remains near the surface for longer periods.

The mapped flood zones show that drainage issues in Hespeler Park are not evenly distributed. Flooding is concentrated in localized low-lying areas where subtle terrain changes, compacted surfaces, pathways, and disconnected drainage features appear to influence water movement. This suggests that many of the drainage problems are connected to local surface conditions rather than one uniform flooding issue across the entire park.

The spatial analysis results support this interpretation by showing that many mapped flood areas occur near one another. Nearest neighbour distances, density mapping, and related spatial summaries show localized clustering of flood-prone areas. These results help show that the inventory is capturing real spatial patterns, but the most important output remains the structured flood-zone database and its ability to support future monitoring.

The observed patterns are consistent with expectations for a low-relief landscape where small depressions, blocked drainage, or insufficient slope can lead to repeated pooling. Pathways, roads, and compacted surfaces within the park may also contribute to redirected flow or drainage interruptions. These conditions help explain why some areas dry seasonally while others remain wet for longer periods.

An aspect-derived surface direction map provided additional support for the terrain interpretation. When compared with the DEM, hillshade, aerial imagery, and mapped flood zones, the aspect layer shows that some flood-prone areas occur where local slope directions converge toward low-lying ground. This supports the interpretation that surface water movement is influenced by subtle changes in elevation that are not always obvious from aerial imagery alone.

The relationship between surface drainage and shallow groundwater is also important. Measured water depths within the Hespeler Wetland range from approximately 5 to 21 cm (see Appendix P), indicating that standing water is maintained close to the ground surface within the system (Town of Niverville, 2021, p. 12). Since runoff from the surrounding park drains toward this wetland, these values provide a useful local reference for considering the limited vertical separation between ground surface elevations and nearby standing water. This does not replace direct groundwater monitoring, but it suggests that deeper excavation or ditching may be constrained in some areas.

Several drainage improvement options may be relevant to the site, including ditch maintenance, culvert or pipe improvements, minor grading, and better connection between existing drainage features. The flood-zone inventory helps identify where these options may deserve closer

review. Larger and more persistent flood zones have a greater effect on park usability, while smaller recurring wet areas may indicate broader surface drainage limitations. This distinction is useful for prioritizing maintenance and deciding where further professional assessment may be needed.

Future work could strengthen this assessment by adding repeated seasonal observations, precipitation records, direct groundwater measurements, and more detailed hydrological modelling. Tools such as SAGA-based hydrological modelling could also be considered if more detailed flow accumulation or drainage modelling is required. Repeating the field mapping after major rainfall or snowmelt events would allow the database to show whether flood zones remain stable, expand, shrink, or respond to maintenance work.

Overall, the project demonstrates that a low-cost GIS workflow can produce a useful flood-zone inventory for a small municipal site. Its strongest contribution is the creation of an organized system for documenting, classifying, and tracking drainage problems over time. This framework can support future monitoring, maintenance planning, comparison of site conditions, and professional drainage review.

RECOMMENDATIONS AND CONCLUSIONS

Given the purpose of this project is to identify flooding issues in Hespeler Park and evaluate possible drainage improvements, conservative recommendations can be made based on the results of the geomatics analysis. Even where elevation differences are minimal, it was observed that many ditches are not connected and that pipes are often inadequate, undersized, or absent. The simplest initial improvement is therefore to review, repair, and connect existing drainage features.

Estimated groundwater proximity was calculated by subtracting an average measured water depth, converted to metres, from LiDAR-derived surface elevation at each field point. Measured water depths were taken from Appendix P and averaged to 0.171 m to represent typical wetland conditions across the study area. Because the wetland holds standing water, the water surface is used as a local reference for estimating vertical separation between the ground surface and nearby standing water conditions. This provides a practical estimate of drainage constraint across the study area (see Appendix R).

Vertical separation is consistently shallow across the study area, generally about 0.17 m. This indicates that drainage improvements in Hespeler Park may be constrained by limited available depth. Excavation deeper than 0.20 m could intersect shallow groundwater or standing water conditions in many areas, limiting the effectiveness of traditional ditching. Drainage interventions should therefore be targeted and conservative. In many areas, ditching may be

less effective than minor grading, shallow surface drainage adjustments, pipe improvements, or better connection of existing drainage features.

Given that the available vertical separation between the ground surface and nearby water levels appears limited, the potential to increase ditch depth is constrained. The proposed routes were selected to follow existing ditches, connect areas that are not currently linked to drainage, and pass through higher trafficked areas such as the baseball fourplex, where flooding has a greater impact on usage. This is intended to improve drainage continuity without unnecessary disturbance to existing structures, except where a larger pipe may be required beneath the paved path. The proposed ditch extension network is intended to connect most major affected areas to existing canals on the north and west of the park (see Appendix S).

The main recommendation is to use the mapped flood-zone inventory as a starting point for maintenance review and professional assessment. The database provides a record of flood locations, severity, likely causes, and priority areas that can be updated after future rainfall, snowmelt, maintenance, or mitigation work. For the next stage, this inventory should be reviewed alongside direct groundwater measurements, seasonal monitoring, precipitation records, and professional hydrological or engineering assessment to determine which drainage improvements are most feasible.

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<https://apps.apple.com/app/id673716080>

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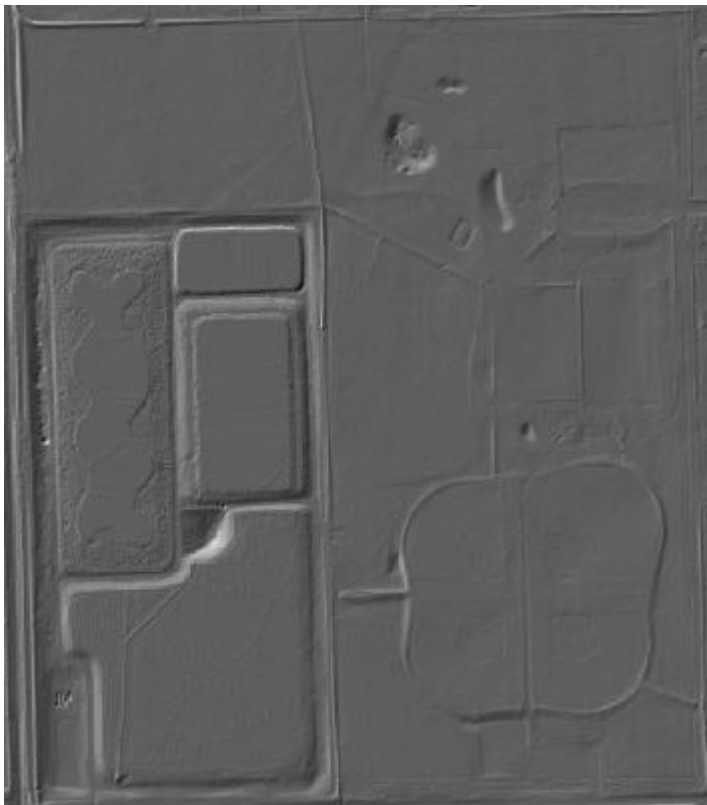
<https://www.epa.gov/wetlands/wetlands-factsheet-series>

APPENDIX

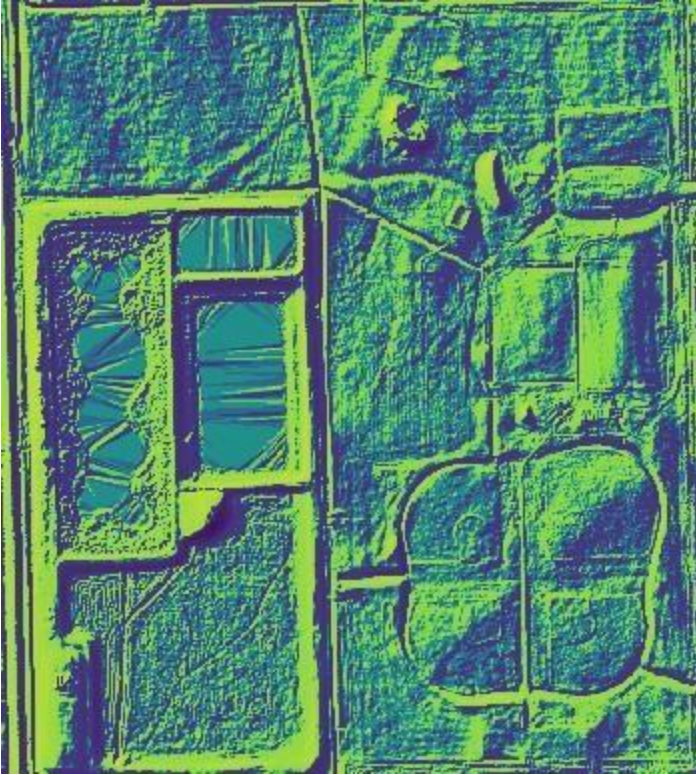
Appendix A: Hespeler Park DEM.



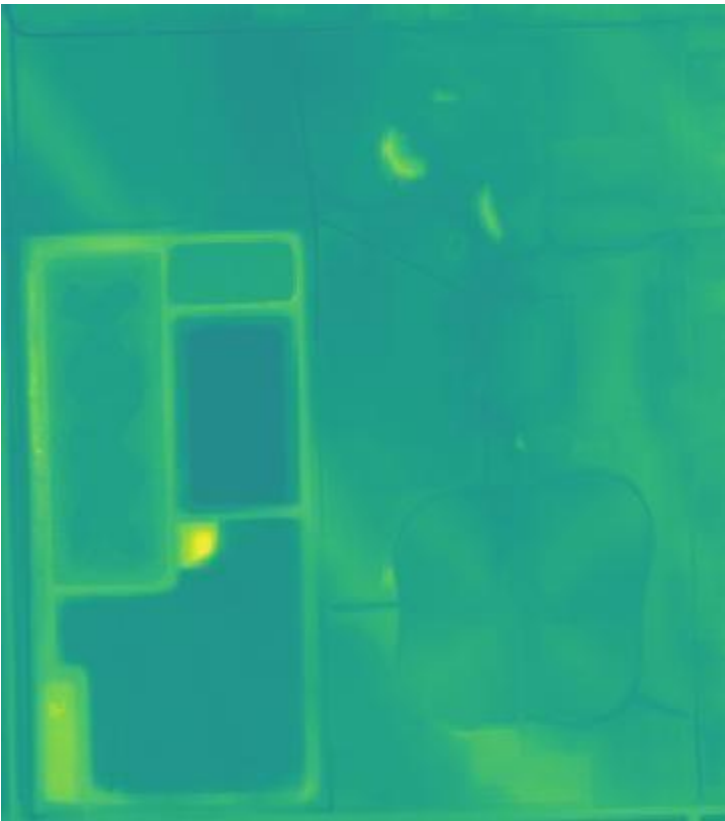
Appendix B: Hespeler Park hillshade Grey.



Appendix C: Hespeler Park hillshade Color.



Appendix D: Hespeler Park DEM color.



Appendix E: Hespeler_Flood_Zones Feature Attributes.

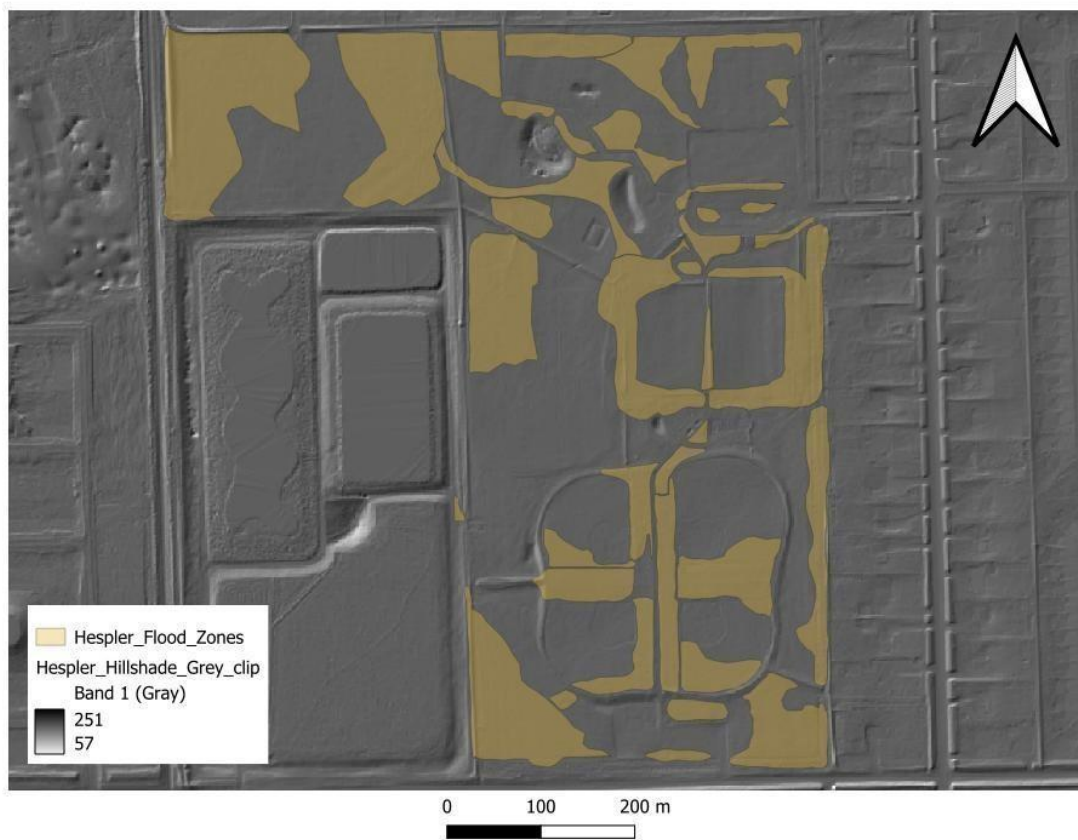
Hespler_Flood_Zones - Feature Attributes

fid	Autogenerate
Flood_ID	NULL
Flood_Tier	(NULL)
Field_Visible	☐
Aerial_Visible	☐
DEM_Visible	☐
Hillshade_Visible	☐
Drainage_Structure	(NULL)
manual_add	☐
shape_edit	☐
Likely_Cause	(NULL)
Priority	(NULL)
Comments	NULL
Confidence	(NULL)

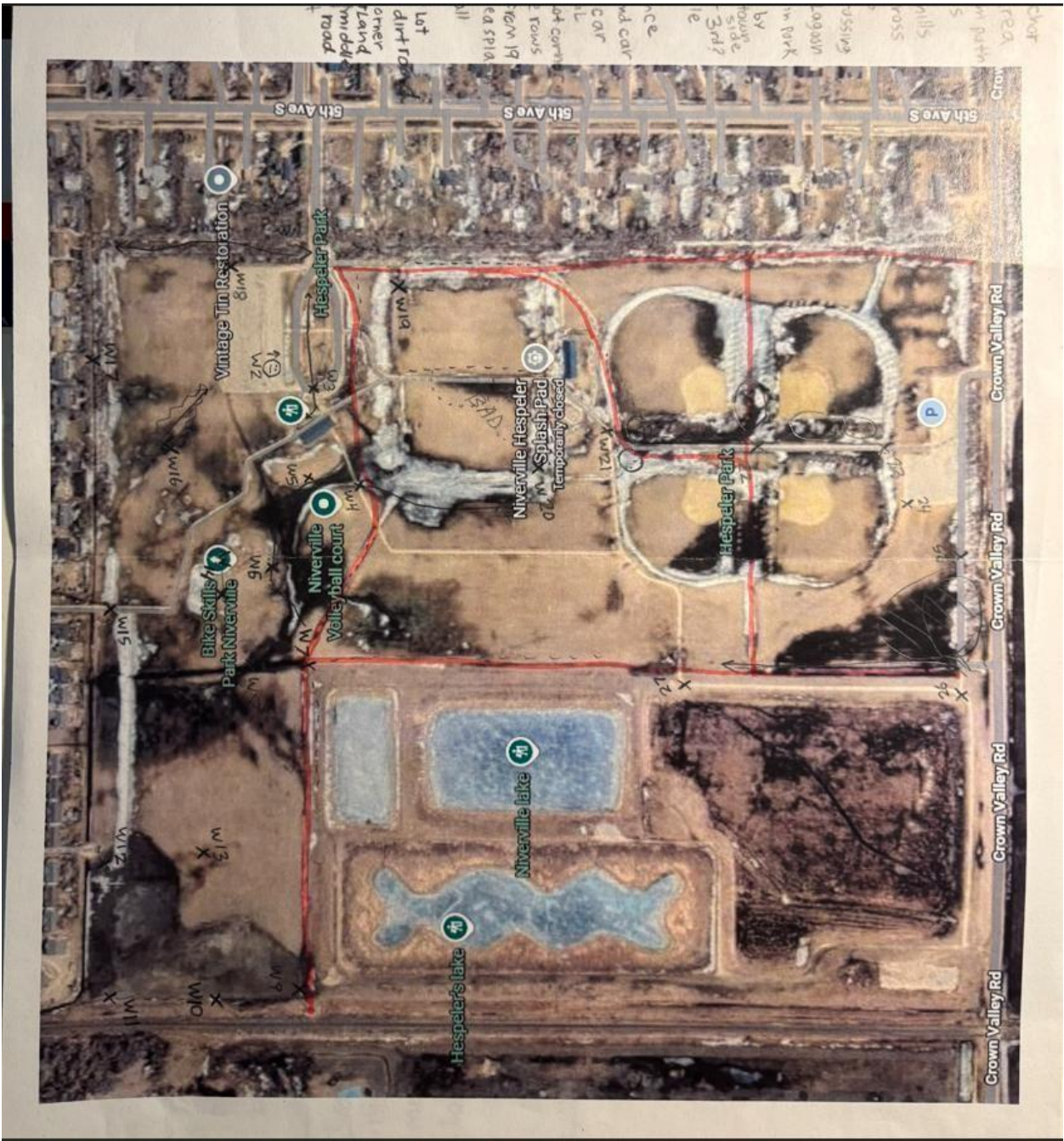
OK Cancel

Appendix F1: Flood Polygon Layer overlaid Hespeler Park.

Flood-Prone Areas in Hespler Park April 2026



Appendix F2: Printout of Hespeler Park used during observation and data collection.



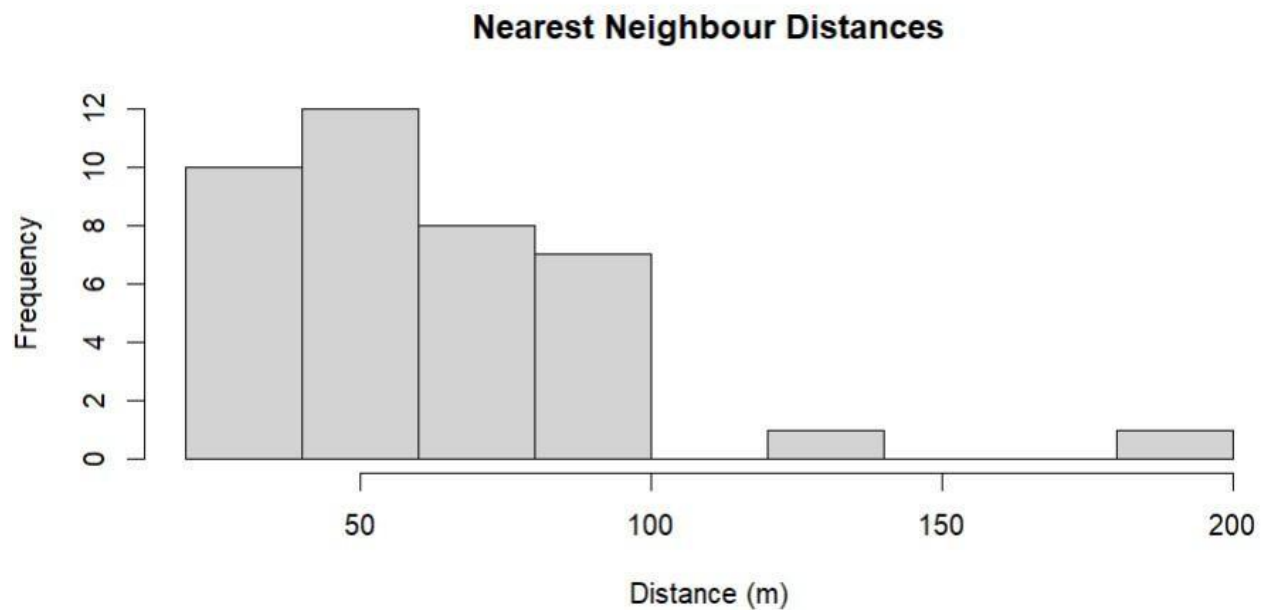
Appendix G: Hespeler Flood Zones Attribute Table.

	fid	Flood_ID	Flood_Tier	Field_Visible	Aerial_Visible	DEM_Visible	hillshade_Visible	ainage_Structu	manual_add	shape_edit	Likely_Cause	Priority	Comments	Confidence
1		27 FZ_27	TIER_3	true	true	true	true	DITCH	true	false	BLOCKED_...	HIGH	ankle deep i...	HIGH
2		34 FZ_34	TIER_3	true	true	true	true	DITCH	true	false	BLOCKED_...	HIGH	ditches clea...	HIGH
3		38 FZ_38	TIER_3	true	true	false	false	DITCH	true	false	POOR_GRA...	MEDIUM	almost entir...	HIGH
4		39 FZ_39	TIER_3	true	true	false	true	MIXED	true	false	POOR_GRA...	HIGH	multiple sm...	HIGH
5		1 FZ_01	TIER_2	true	false	false	true	PIPE	true	false	POOR_GRA...	MEDIUM	water has n...	HIGH
6		2 FZ_02	TIER_2	true	true	false	false	NONE	true	false	BLOCKED_...	MEDIUM	low point cr...	HIGH
7		3 FZ_03	TIER_2	true	true	true	false	NONE	true	false	BLOCKED_...	MEDIUM	paths and r...	HIGH
8		4 FZ_04	TIER_2	true	true	false	false	NONE	true	false	BLOCKED_...	MEDIUM	blocked fro...	HIGH
9		5 FZ_05	TIER_2	true	true	false	false	NONE	true	false	POOR_GRA...	MEDIUM	<i>NULL</i>	HIGH
10		6 FZ_06	TIER_2	true	true	true	true	DITCH	true	false	POOR_GRA...	HIGH	it covers on...	HIGH
11		7 FZ_07	TIER_2	true	true	true	false	NONE	true	false	BLOCKED_...	MEDIUM	no way to d...	HIGH
12		9 FZ_09	TIER_2	true	true	true	true	DITCH	true	false	POOR_GRA...	HIGH	restricts use...	HIGH
13		10 FZ_10	TIER_2	true	true	true	true	DITCH	true	false	POOR_GRA...	HIGH	water pools...	HIGH
14		11 FZ_11	TIER_2	true	true	false	false	NONE	true	false	POOR_GRA...	LOW	beside a ditch	HIGH
15		12 FZ_12	TIER_2	true	true	true	true	DITCH	true	false	POOR_GRA...	HIGH	large portio...	HIGH
16		13 FZ_13	TIER_2	true	true	false	true	DITCH	true	false	POOR_GRA...	MEDIUM	<i>NULL</i>	HIGH
17		15 FZ_15	TIER_2	true	true	false	false	DITCH	true	false	POOR_GRA...	MEDIUM	<i>NULL</i>	HIGH
18		18 FZ_18	TIER_2	true	true	false	false	DITCH	true	false	POOR_GRA...	HIGH	Middle of hi...	HIGH
19		19 FZ_19	TIER_2	true	true	false	false	NONE	true	false	POOR_GRA...	MEDIUM	High use area	HIGH
20		20 FZ_17	TIER_2	true	true	false	false	NONE	true	false	POOR_GRA...	MEDIUM	<i>NULL</i>	HIGH
21		23 FZ_21	TIER_2	true	true	false	false	NONE	true	false	BLOCKED_...	HIGH	small childr...	HIGH
22		24 FZ_22	TIER_2	true	true	false	true	DITCH	true	false	BLOCKED_...	HIGH	path is only...	HIGH
23		25 FZ_25	TIER_2	true	true	false	false	MIXED	true	false	POOR_GRA...	HIGH	manholes a...	HIGH
24		26 FZ_26	TIER_2	true	true	false	false	DITCH	true	false	POOR_GRA...	LOW	<i>NULL</i>	HIGH
25		29 FZ_28	TIER_2	true	true	true	true	DITCH	true	false	SNOW_POOL	MEDIUM	water trapp...	HIGH
26		30 FZ_29	TIER_2	true	true	true	true	NONE	true	false	BLOCKED_...	MEDIUM	nowhere for...	HIGH
27		31 FZ_30	TIER_2	true	true	false	true	DITCH	true	false	POOR_GRA...	MEDIUM	large area o...	HIGH
28		33 FZ_33	TIER_2	true	true	true	true	DITCH	true	false	BLOCKED_...	HIGH	persistantly ...	HIGH
29		35 FZ_35	TIER_2	true	true	false	false	DITCH	true	false	SNOW_POOL	LOW	<i>NULL</i>	HIGH
30		36 FZ_36	TIER_2	true	true	false	true	DITCH	true	false	BLOCKED_...	HIGH	nowhere to ...	HIGH

Appendix H: Summary statistics of flood zones by tier.

	Flood_Tier	count	total_area_m2	mean_area_m2	min_area_m2	max_area_m2
	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	TIER_1	8	10218.	1277.	136.	3518.
2	TIER_2	27	113122.	4190.	193.	20527.
3	TIER_3	4	20942.	5235.	2545.	6821.
>						

Appendix I: Nearest Neighbour Distance Distribution.



Appendix J: Moran's I Output.

Moran I test under randomisation

data: flood\$Flood_Tier_num

weights: lw

n reduced by no-neighbour observations

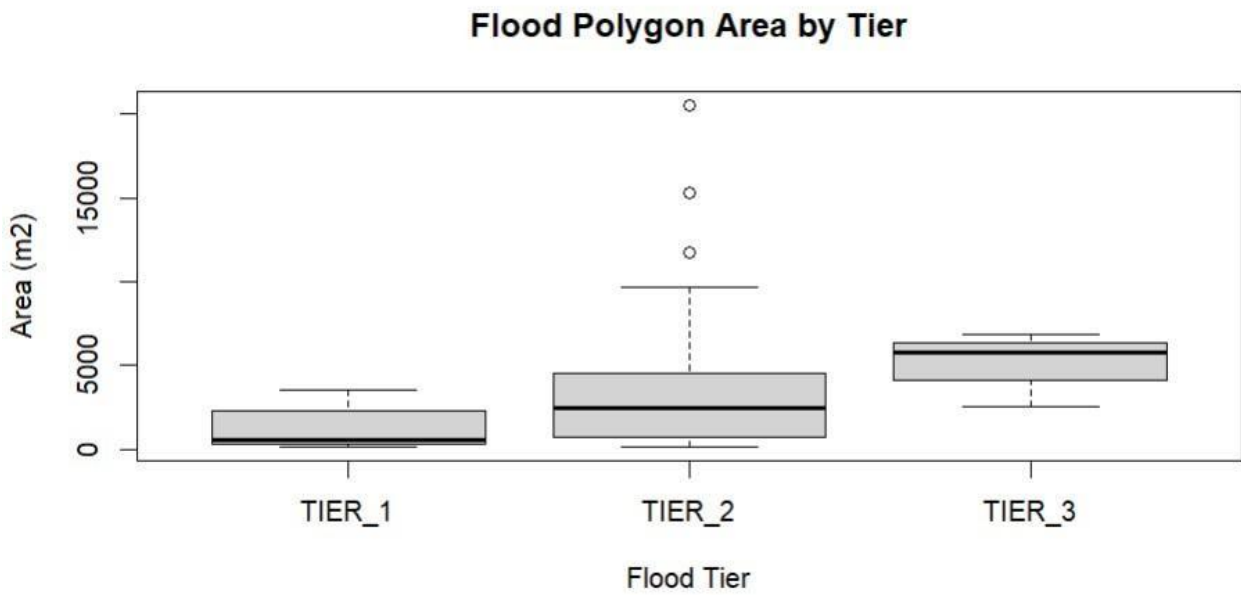
Moran I statistic standard deviate = 1.1605, p-value = 0.1229

alternative hypothesis: greater

sample estimates:

Moran I statistic	Expectation	Variance
0.13000624	-0.02777778	0.01848420

Appendix K: Flood Polygon Area by Tier.



Appendix L: Flood Priority Counts.

HIGH	LOW	MEDIUM
13	8	18

Appendix M1,2,3,4: Complete RStudio code used in project.

M1:

Appendix:M1

```
# Read GeoPackage and flood polygon layer
flood <- st_read("Hespler_Project.gpkg", layer = "Hespler_Flood_Zones")

# Repair geometry and calculate polygon area
flood <- st_make_valid(flood)
flood$area_m2 <- as.numeric(st_area(flood))

# Summary statistics by flood tier
flood %>%
  st_drop_geometry() %>%
  group_by(Flood_Tier) %>%
  summarise(
    count = n(),
    total_area_m2 = sum(area_m2, na.rm = TRUE),
    mean_area_m2 = mean(area_m2, na.rm = TRUE),
    min_area_m2 = min(area_m2, na.rm = TRUE),
    max_area_m2 = max(area_m2, na.rm = TRUE)
  )

# Centroids for spatial pattern analysis
centroids <- st_centroid(flood)

# Nearest neighbour distances
coords <- st_coordinates(centroids)
dist_matrix <- as.matrix(dist(coords))
diag(dist_matrix) <- NA
nn_dist <- apply(dist_matrix, 1, min, na.rm = TRUE)

# Nearest neighbour histogram
hist(nn_dist,
      main = "Nearest Neighbour Distances",
      xlab = "Distance (m)")
```

M2:

```

Appendix:M2
# Nearest neighbour summary statistics
summary(nn_dist)
mean(nn_dist)
median(nn_dist)
sd(nn_dist)

# Moran's I using flood tier values
nb <- dnearneigh(coords, 0, 100)
lw <- nb2listw(nb, style = "w", zero.policy = TRUE)
flood$Flood_Tier_num <- as.numeric(gsub("TIER_", "", flood$Flood_Tier))
moran.test(flood$Flood_Tier_num, lw, zero.policy = TRUE)

# Flood polygon area by tier
boxplot(area_m2 ~ Flood_Tier,
        data = st_drop_geometry(flood),
        main = "Flood Polygon Area by Tier",
        xlab = "Flood Tier",
        ylab = "Area (m2)")

# Flood priority counts
table(flood$Priority)

```

M3:

```

centroids <- st_centroid(flood)

coords <- st_coordinates(centroids)

pp <- ppp(coords[,1], coords[,2],
          window = owin(range(coords[,1]), range(coords[,2])))

kd <- density(pp, sigma = 50)

plot(kd)

```

M4:

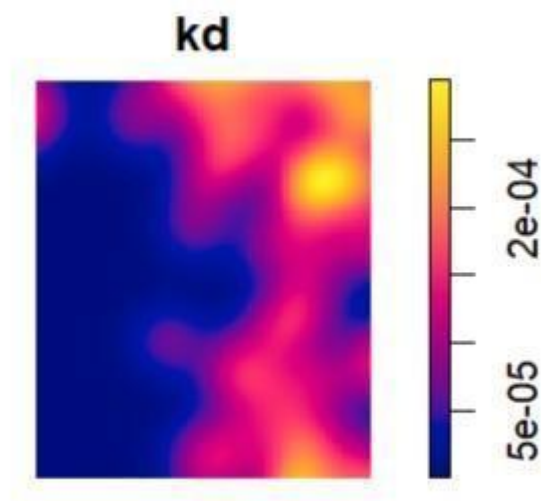
```
coords <- st_coordinates(centroids)

pp <- ppp(coords[,1], coords[,2],
          window = owin(range(coords[,1]), range(coords[,2])))

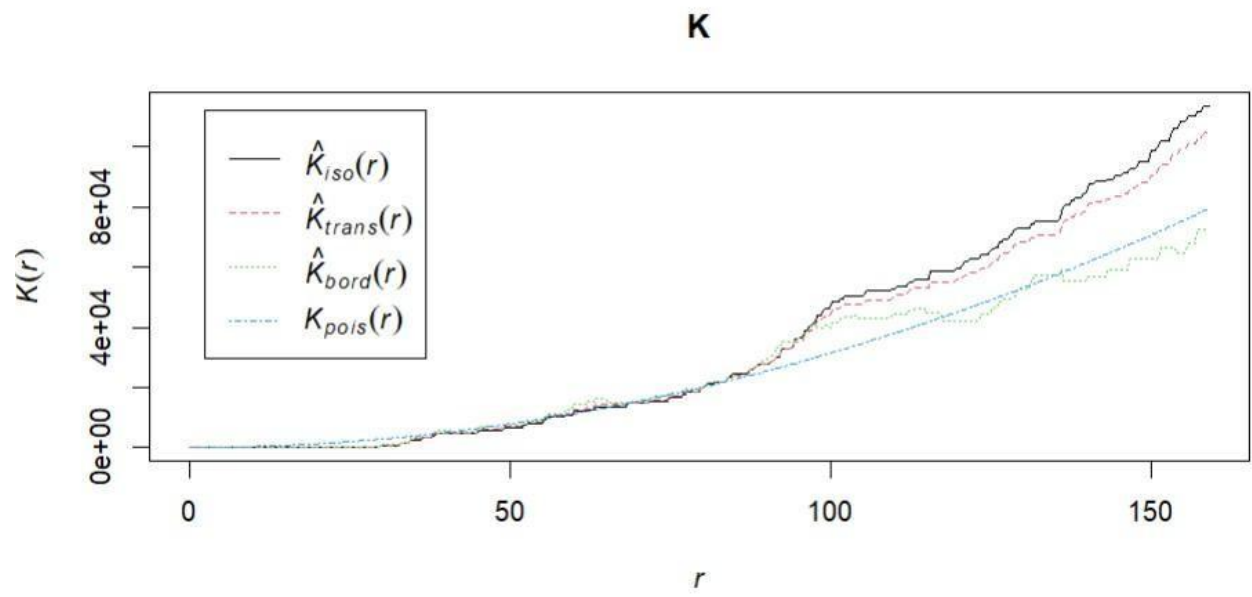
K <- Kest(pp)

plot(K)
```

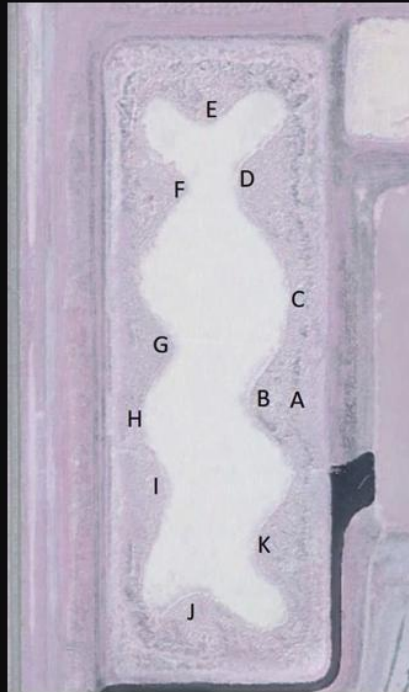
Appendix N: Kernel Density of Flood Zone Centroids.



Appendix O: Ripley's K-function Output.



Appendix P: Wetland Project water levels.

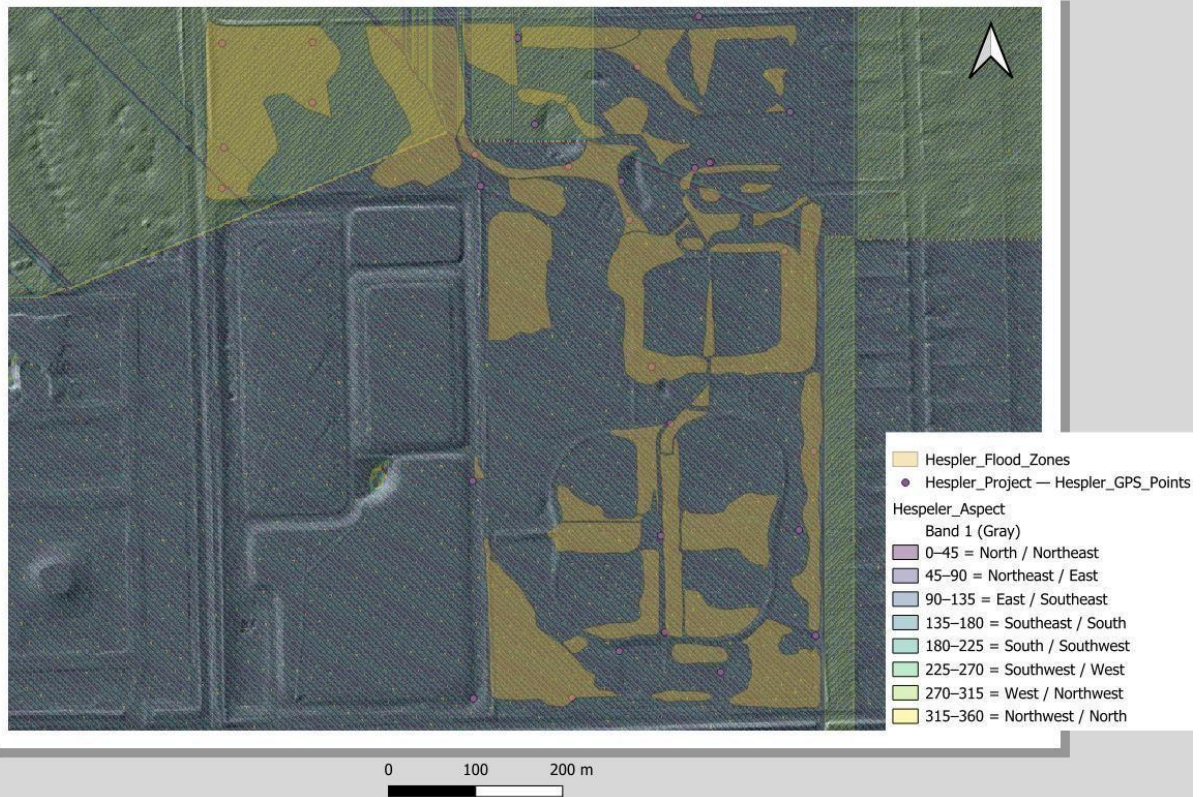


Location	Water depth (cm)
A	5
B	18
C	17
D	21
E	20
F	19
G	14
H	18
I	15
J	20
K	21

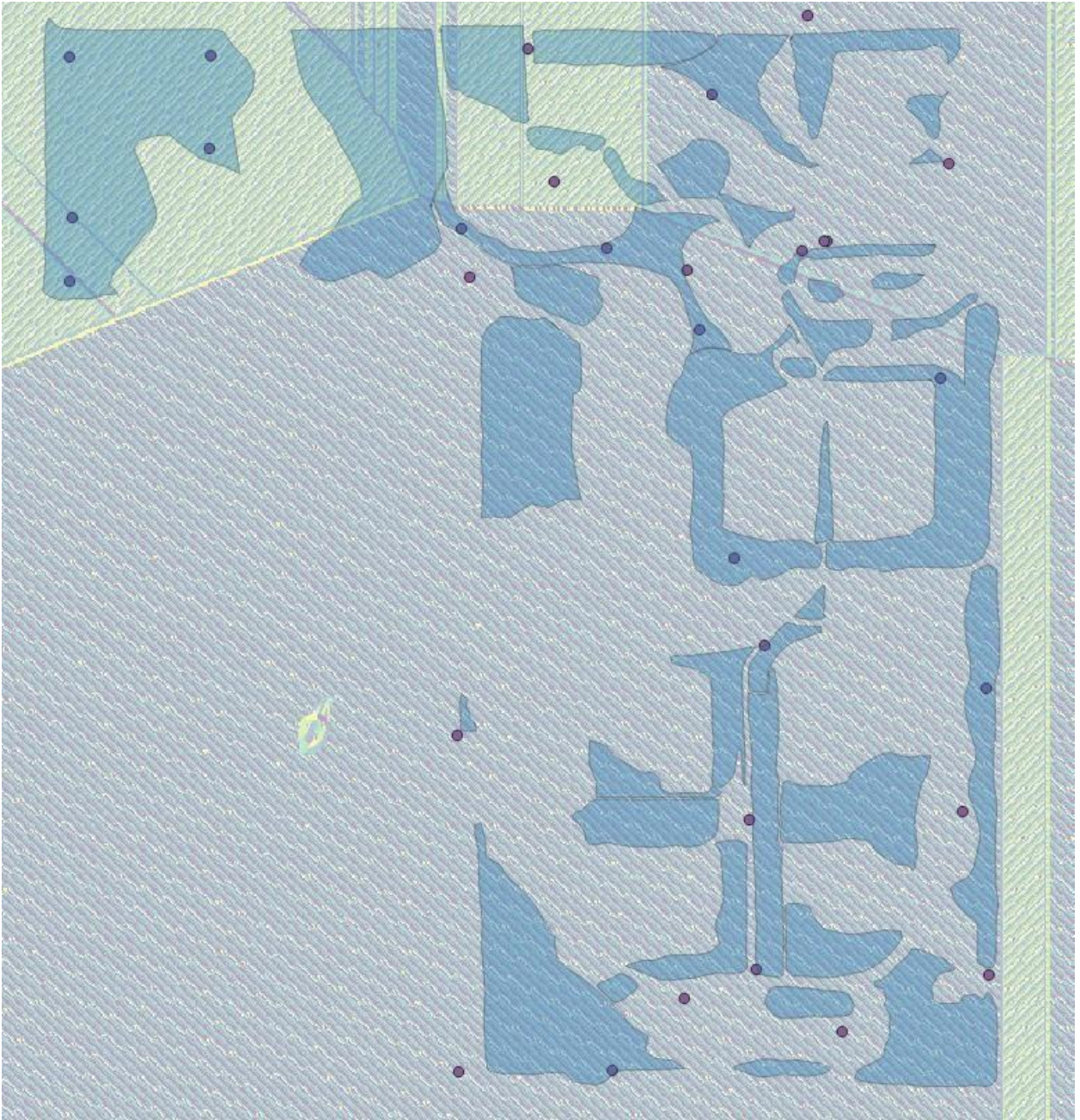
Figure 10. Wetland bench water levels, August 10, 2020.

Appendix Q1: Aspect derived surface direction of flow with hillshade layer.

Aspect-Derived Surface Direction and Flood Zones in Hespeler Park April 2026



Q2: Aspect derived surface direction of flow with GPS points.



Appendix R: Vertical separation between averaged local ground water and ground elevation.

Appendix R: Estimated Groundwater Elevation and Separation

Waypoint	Surface elevation (m)	Avg. depth used (m)	Estimated groundwater elevation (m)
31	237.4	0.171	237.229
30	234.9	0.171	234.729
29	235.8	0.171	235.629
28	238.6	0.171	238.429
27	240.6	0.171	240.429
26	240.2	0.171	240.029
25	236	0.171	235.829
24	238.7	0.171	238.529
23	235.2	0.171	235.029
22	236	0.171	235.829
21	239.5	0.171	239.329
20	237.2	0.171	237.029
19	235.5	0.171	235.329
18	236.7	0.171	236.529
17	239.4	0.171	239.229
16	239.6	0.171	239.429
15	237.5	0.171	237.329
14	238.3	0.171	238.129
13	237.1	0.171	236.929
12	236.4	0.171	236.229
11	238.6	0.171	238.429
10	238.7	0.171	238.529
9	239.1	0.171	238.929
8	236.5	0.171	236.329
7	236.1	0.171	235.929
6	239.2	0.171	239.029
5	241.3	0.171	241.129
4	238.4	0.171	238.229
3	237.8	0.171	237.629
2	241.4	0.171	241.229
1	237.1	0.171	236.929

Appendix S: Recommended ditch system to increase drainage across highest trafficked areas.

Proposed Shallow Ditch Network

